

Wireshark Packet Analysis Project

1. HTTP GET Traffic Analysis

Using *http://neverssl.com* — a site that deliberately stays on HTTP — filtered it in Wireshark with:

```
http.request.method == "GET"
```

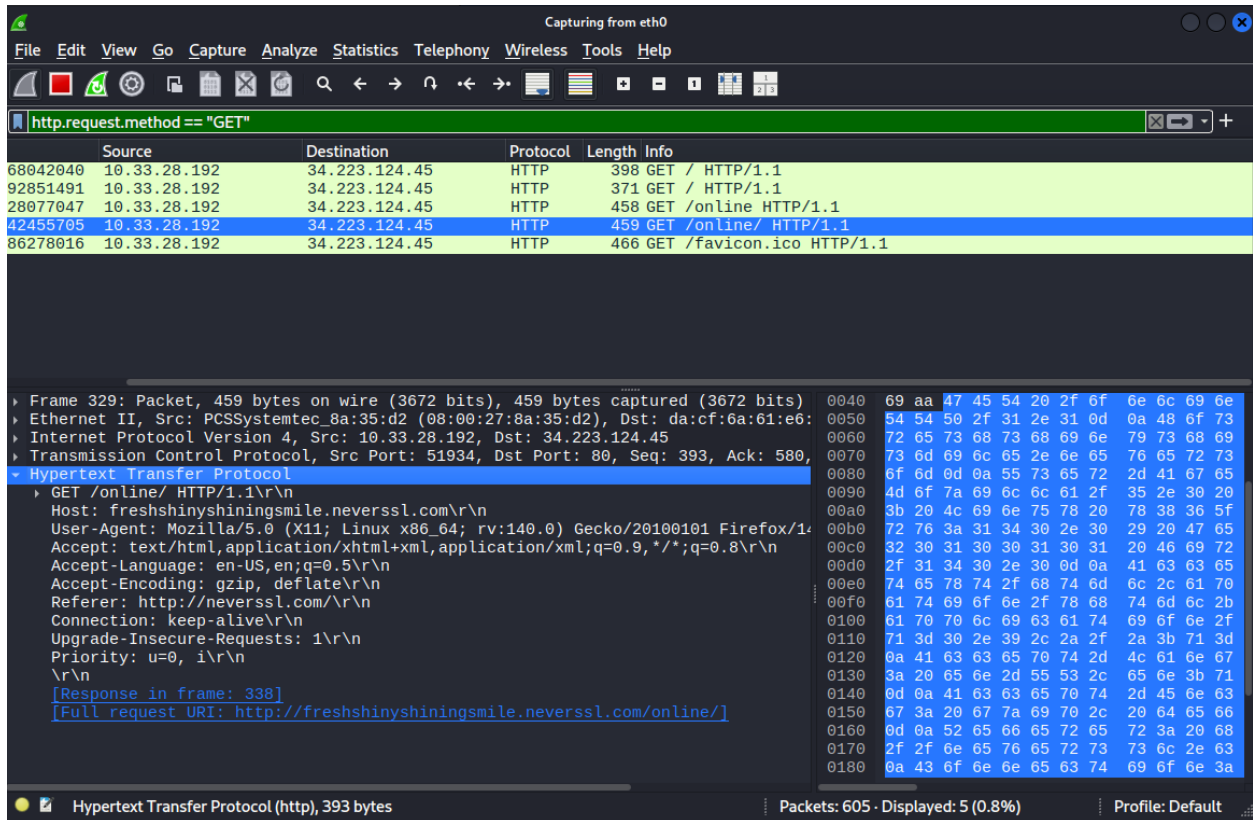
What the capture shows

Source IP	10.33.28.192 (my machine)
Destination IP	34.223.124.45
Port	80 (standard HTTP)
Requests captured	5 GET requests
Paths requested	/, /online, /online/, /favicon.ico
Host header	freshshinyshiningmile.neverssl.com

Analysis

The selected packet — GET /online/ HTTP/1.1 — exposes everything: the User-Agent string, Accept-Language headers, Referer, and connection settings, all in plain text. There's no encryption here. Anyone on the same network could read every header and the full response body without needing to crack anything.

This is why browsers now flag HTTP sites as 'Not Secure'. The packet makes it obvious why.



2. DNS Query and Response

DNS runs on UDP port 53 and handles every domain lookup before a connection opens. I applied the filter:

dns

DNS Query — Packet 274

Source	10.33.28.192 (my machine)
Destination	10.33.28.34 (DNS server)
Protocol	UDP, port 53
Transaction ID	0x8bbd
Type	Standard Query
Domain queried	freshshinyshiningmile.neverssl.com
Answer RRs	0 — query only, no answer yet

The image shows a Wireshark packet capture window titled "Capturing from eth0". The filter is set to "dns". The packet list shows several DNS queries and responses. Packet 274 is selected, showing a DNS response from source 10.33.28.34 to destination 10.33.28.192. The response details show a transaction ID of 0x8bbd, flags of 0x0100 (Standard query response), and one answer RR: 192.168.1.1. The packet bytes pane shows the raw data of the response.

No.	Time	Source	Destination	Protocol	Length	Info
259	54.178264117	10.33.28.192	10.33.28.34	DNS	95	Standard query 0x835f AAAA freshshinyshiningsmil
260	54.205690788	10.33.28.34	10.33.28.192	DNS	111	Standard query response 0x975a A freshshinyshini
261	54.205691861	10.33.28.34	10.33.28.192	DNS	123	Standard query response 0x835f AAAA freshshinysh
273	55.477266208	10.33.28.192	10.33.28.34	DNS	95	Standard query 0x12bf A freshshinyshiningsmile.n
274	55.477406414	10.33.28.192	10.33.28.34	DNS	95	Standard query 0x8bbd AAAA freshshinyshiningsmil
275	55.483014590	10.33.28.34	10.33.28.192	DNS	111	Standard query response 0x12bf A freshshinyshini
276	55.483015202	10.33.28.34	10.33.28.192	DNS	123	Standard query response 0x8bbd AAAA freshshinysh
621	233.168815519	10.33.28.192	10.33.28.34	DNS	70	Standard query 0xa663 A google.com
622	233.238759640	10.33.28.34	10.33.28.192	DNS	86	Standard query response 0xa663 A google.com A 14
623	233.239740591	10.33.28.192	10.33.28.34	DNS	70	Standard query 0x1dd4 AAAA google.com
624	233.297783490	10.33.28.34	10.33.28.192	DNS	98	Standard query response 0x1dd4 AAAA google.com A
631	273.925645768	10.33.28.192	10.33.28.34	DNS	75	Standard query 0x483a A ads.mozilla.org
632	274.008812861	10.33.28.192	10.33.28.34	DNS	72	Standard query 0x539c A neverssl.com

Frame 274: Packet, 95 bytes on wire (760 bits), 95 bytes captured (760 bits) on interface eth0
 Ethernet II, Src: PCSSystemtec_8a:35:d2 (08:00:27:8a:35:d2), Dst: da:cf:6a:61:e6:1c
 Internet Protocol Version 4, Src: 10.33.28.192, Dst: 10.33.28.34
 User Datagram Protocol, Src Port: 41967, Dst Port: 53
 Source Port: 41967
 Destination Port: 53
 Length: 61
 Checksum: 0x4d72 [unverified]
 [Checksum Status: Unverified]
 [Stream index: 13]
 [Stream Packet Number: 2]
 [Timestamps]
 UDP payload (53 bytes)
 Domain Name System (query)
 Transaction ID: 0x8bbd
 Flags: 0x0100 Standard query response
 Questions: 1
 Answer RRs: 0
 Authority RRs: 0
 Additional RRs: 0
 Queries
 [Response In: 276]

Domain Name System: Protocol Packets: 691 · Displayed: 56 (8.1%) Profile: Default

DNS Response — Packet 276

Source	10.33.28.34 (DNS server)
Destination	10.33.28.192 (my machine)
Transaction ID	0x8bbd (matches query — same exchange)
Flags	Standard query response, No error
Answer RRs	1 — IP address returned
Resolution time	5.61 ms

Capturing from eth0

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dns

No.	Time	Source	Destination	Protocol	Length	Info
259	54.178264117	10.33.28.192	10.33.28.34	DNS	95	Standard query 0x835f AAAA freshshinyshiningsmil
260	54.205690788	10.33.28.34	10.33.28.192	DNS	111	Standard query response 0x975a A freshshinyshini
261	54.205691861	10.33.28.34	10.33.28.192	DNS	123	Standard query response 0x835f AAAA freshshinysh
273	55.477266208	10.33.28.192	10.33.28.34	DNS	95	Standard query 0x12bf A freshshinyshiningsmile.n
274	55.477406414	10.33.28.192	10.33.28.34	DNS	95	Standard query 0x8bbd AAAA freshshinyshiningsmil
275	55.483014590	10.33.28.34	10.33.28.192	DNS	111	Standard query response 0x12bf A freshshinyshini
276	55.483015202	10.33.28.34	10.33.28.192	DNS	123	Standard query response 0x8bbd AAAA freshshinysh
621	233.168815519	10.33.28.192	10.33.28.34	DNS	70	Standard query 0xa663 A google.com
622	233.238759640	10.33.28.34	10.33.28.192	DNS	86	Standard query response 0xa663 A google.com A 14
623	233.239740591	10.33.28.192	10.33.28.34	DNS	70	Standard query 0x1dd4 AAAA google.com
624	233.297783490	10.33.28.34	10.33.28.192	DNS	98	Standard query response 0x1dd4 AAAA google.com A
631	273.925645768	10.33.28.192	10.33.28.34	DNS	75	Standard query 0x483a A ads.mozilla.org
632	274.008812861	10.33.28.192	10.33.28.34	DNS	72	Standard query 0x539c A neverssl.com
633	274.009120629	10.33.28.192	10.33.28.34	DNS	72	Standard query 0x7c9a AAAA neverssl.com

Frame 276: Packet, 123 bytes on wire (984 bits), 123 bytes captured (984 bits) on interface eth0
 Ethernet II, Src: da:cf:6a:61:e6:0a (da:cf:6a:61:e6:0a), Dst: PCSSystemtec_8a:35:00:00:00:00
 Internet Protocol Version 4, Src: 10.33.28.34, Dst: 10.33.28.192
 User Datagram Protocol, Src Port: 53, Dst Port: 41967
 Source Port: 53
 Destination Port: 41967
 Length: 89
 Checksum: 0xd82 [unverified]
 [Checksum Status: Unverified]
 [Stream index: 13]
 [Stream Packet Number: 4]
 [Timestamps]
 UDP payload (81 bytes)
 Domain Name System (response)
 Transaction ID: 0x8bbd
 Flags: 0x8180 Standard query response, No error
 Questions: 1
 Answer RRs: 1
 Authority RRs: 0
 Additional RRs: 0
 Queries
 Answers
 [Request In: 274]
 [Time: 5.608788 milliseconds]

Domain Name System: Protocol Packets: 1845 - Displayed: 74 (4.0%) Profile: Default

Analysis

The matching Transaction IDs (0x8bbd on both sides) confirm this is one paired exchange. The client asked; the server answered 5.6ms later with an IP address the browser could actually connect to.

What's worth noting: DNS is unencrypted by default. Every domain your machine looks up is visible on the network. That's why DNS-over-HTTPS (DoH) exists — but standard DNS, as shown here, leaks your browsing destinations even when the actual traffic is encrypted.

The capture also shows several other lookups happening in the background — google.com, ads.mozilla.org, neverssl.com — which is a reminder of how many DNS requests a browser makes that users never see.

3. TLS Handshake — Client Hello and Server Hello

Visiting <https://google.com> triggers a TLS handshake before any application data moves. The filter used:

```
tls.handshake.type == 1 || tls.handshake.type == 2
```

Client Hello — Packet 2931

Source	10.33.28.192 (my machine)
Destination	172.217.22.106 (Google)
Port	443 (HTTPS)
TLS Record Version	TLS 1.0 (0x0301) — used for compatibility
Handshake Version	TLS 1.2 (0x0303)
Cipher suites offered	17 suites
SNI	ogads-pa.clients6.google.com
Supported versions ext.	TLS 1.3 and TLS 1.2
Key share	X25519MLKEM768, x25519, secp256r1

Capturing from eth0

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tls

No.	Time	Source	Destination	Protocol	Length	Info
2915	390.644624299	172.217.22.106	10.33.28.192	QUIC	1294	Initial, DCID=d30830, SCID=ea261a88a65fd796, PKN
2931	390.676654924	10.33.28.192	172.217.22.106	TLSv1.3	1979	Client Hello (SNI=ogads-pa.clients6.google.com)
2959	390.853067879	172.217.22.106	10.33.28.192	TLSv1.3	2842	Server Hello, Change Cipher Spec
2967	390.864978182	172.217.22.106	10.33.28.192	TLSv1.3	1109	Application Data
2988	391.072344287	10.33.28.192	172.217.22.106	TLSv1.3	146	Change Cipher Spec, Application Data
2989	391.073018182	10.33.28.192	172.217.22.106	TLSv1.3	158	Application Data
3004	391.228809889	172.217.22.106	10.33.28.192	TLSv1.3	716	Application Data, Application Data
3006	391.229965680	10.33.28.192	172.217.22.106	TLSv1.3	97	Application Data
3007	391.232946585	172.217.22.106	10.33.28.192	TLSv1.3	97	Application Data
3028	391.649260910	10.33.28.192	142.250.140.138	TLSv1.3	1966	Client Hello (SNI=play.google.com)
3029	391.650198958	10.33.28.192	142.250.140.138	TLSv1.3	1966	Client Hello (SNI=play.google.com)

Frame 2931: Packet, 1979 bytes on wire (15832 bits), 1979 bytes captured (15832 b...
 Ethernet II, Src: PCSSystemtec.8a:35:d2 (08:00:27:8a:35:d2), Dst: da:cf:6a:61:e6:
 Internet Protocol Version 4, Src: 10.33.28.192, Dst: 172.217.22.106
 Transmission Control Protocol, Src Port: 50980, Dst Port: 443, Seq: 1, Ack: 1, Le
 Transport Layer Security
 [Stream index: 28]
 TLSv1.3 Record Layer: Handshake Protocol: Client Hello
 Content Type: Handshake (22)
 Version: TLS 1.0 (0x0301)
 Length: 1908
 Handshake Protocol: Client Hello
 Handshake Type: Client Hello (1)
 Length: 1904
 Version: TLS 1.2 (0x0303)
 Random: 72be8897bd871c725d0e67a8fea9e1bfd345f64130ef1492f5a141cc95180d35
 Session ID Length: 32
 Session ID: cc8a12a966ed097bd8e64df5f39c7d3c0f544a8bbbaabe5045e627f388a0ad
 Cipher Suites Length: 34
 Cipher Suites (17 suites)
 Compression Methods Length: 1
 Compression Methods (1 method)
 Extensions Length: 1797
 Extension: server_name (len=33) name=ogads-pa.clients6.google.com
 Extension: extended_master_secret (len=0)
 Extension: renegotiation_info (len=1)
 Extension: supported_groups (len=16)
 Extension: ec_point_formats (len=2)
 Extension: session_ticket (len=0)
 Extension: application_layer_protocol_negotiation (len=14)
 Extension: status_request (len=5)
 Extension: delegated_credentials (len=10)
 Extension: signed_certificate_timestamp (len=0)
 Extension: key_share (len=1327) X25519MLKEM768, x25519, secp256r1
 Extension: supported_versions (len=5) TLS 1.3, TLS 1.2
 Extension: signature_algorithms (len=24)
 Extension: psk_key_exchange_modes (len=2)
 Extension: record_size_limit (len=2)
 Extension: compress_certificate (len=7)
 Extension: encrypted_client_hello (len=281)
 [JA4: t13d1717h2_5b57614c22b0_3cbfd9057e0d]
 [JA4_r: t13d1717h2_002f,0035,009c,009d,1301,1302,1303,c009,c00a,c013,c014,
 [JA3 Fullstring: 771,4865-4867-4866-49195-49199-52393-52392-49196-49200-49
 [JA3: 6f7889b9fb1a62a9577e685c1fcfa919]

Transport Layer Security: Protocol Packets: 3271 · Displayed: 634 (19.4%) Profile: Default

Server Hello — Packet 2959

Source	172.217.22.106 (Google)
Destination	10.33.28.192 (my machine)
Cipher suite selected	TLS_AES_256_GCM_SHA384 (0x1302)
TLS version confirmed	TLS 1.2 (0x0303) record layer
Session ID	Matches Client Hello (same session)
Compression method	null (none)
Followed by	Change Cipher Spec, then encrypted Application Data

Capturing from eth0

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tls

No.	Time	Source	Destination	Protocol	Length	Info
2915	390.644624299	172.217.22.106	10.33.28.192	QUIC	1294	Initial, DCID=d30830, SCID=ea261a88a65fd796, PKN
2931	390.676654924	10.33.28.192	172.217.22.106	TLSv1.3	1979	Client Hello (SNI=ogads-pa.clients6.google.com)
2959	390.853067879	172.217.22.106	10.33.28.192	TLSv1.3	2842	Server Hello, Change Cipher Spec
2967	390.864978182	172.217.22.106	10.33.28.192	TLSv1.3	1109	Application Data
2988	391.072344287	10.33.28.192	172.217.22.106	TLSv1.3	146	Change Cipher Spec, Application Data
2989	391.073018182	10.33.28.192	172.217.22.106	TLSv1.3	158	Application Data
3004	391.228809889	172.217.22.106	10.33.28.192	TLSv1.3	716	Application Data, Application Data
3006	391.229965680	10.33.28.192	172.217.22.106	TLSv1.3	97	Application Data

Frame 2959: Packet, 2842 bytes on wire (22736 bits), 2842 bytes captured (22736 b

Ethernet II, Src: da:cf:6a:61:e6:0a (da:cf:6a:61:e6:0a), Dst: PCSSystemtec_Ba:35:

Internet Protocol Version 4, Src: 172.217.22.106, Dst: 10.33.28.192

Transmission Control Protocol, Src Port: 443, Dst Port: 50980, Seq: 1, Ack: 1914,

Transport Layer Security

[Stream index: 28]

TLSv1.3 Record Layer: Handshake Protocol: Server Hello

Content Type: Handshake (22)

Version: TLS 1.2 (0x0303)

Length: 1210

Handshake Protocol: Server Hello

Handshake Type: Server Hello (2)

Length: 1206

Version: TLS 1.2 (0x0303)

Random: e6c13f9f1ee6b025bb907d4240d35102eae45adc1b3ec4822fbbd2af3a44efc4

Session ID Length: 32

Session ID: cc8a12a966ed097bd8e64df5f39c7d3c0f544a8bbbbaabe5045e627f388a0ad

Cipher Suite: TLS_AES_256_GCM_SHA384 (0x1302)

Compression Method: null (0)

Extensions Length: 1134

Extension: key_share (len=1124) X25519MLKEM768

Extension: supported_versions (len=2) TLS 1.3

[JA3S: Fullstring: 771,4866,51-43]

[JA3S: 907bf3cecf1c987c889946b737b43de8]

TLSv1.3 Record Layer: Change Cipher Spec Protocol: Change Cipher Spec

Content Type: Change Cipher Spec (20)

Version: TLS 1.2 (0x0303)

Length: 1

Change Cipher Spec Message

TLS segment data (1555 bytes)

0000 08 00 27 8a 35 d2 da cf 6a 61 e6 0a

0010 0b 0c 0d 9e 00 00 78 06 40 2a ac d9

0020 1c c0 01 bb c7 24 73 cb b5 8d 4f 16

0030 04 13 f5 22 00 00 01 01 08 0a 9e 16

0040 41 42 16 03 03 04 ba 02 00 04 b6 03

0050 9f 1e e6 b0 25 bb 90 7d 42 40 d3 51

0060 dc 1b 3e c4 82 2f bd d2 af 3a 44 ef

0070 12 a9 66 ed 09 7b d8 e6 4d f5 f3 9c

0080 4a 8b bb aa be 50 45 e6 27 f3 88 a0

0090 00 04 6e 00 33 04 64 11 ec 04 60 ad

00a0 6c 84 06 38 85 06 26 59 cb 0c 42 50

00b0 e1 81 bc 2f 9c 4e bc bf a2 07 de 8e

00c0 e0 9e 27 b3 3f 4a 0b 17 45 e4 84 31

00d0 6f 47 9c b8 9d 61 77 f7 b4 cf fc d6

00e0 e7 e7 6c 11 d6 c3 7e 98 3b 0d 40 e0

00f0 0c 26 e0 d7 29 1a 62 90 c1 aa ba 46

0100 a2 36 e6 ac 2c 5b 5e c4 4d 29 b7 46

0110 e7 29 64 9f 26 17 9f f2 e5 c2 ed 07

0120 5e 12 44 c1 b5 3e 2a cf d8 70 89 be

0130 0d 65 c8 cc 6f 7f 45 ab 23 ed 83 a8

0140 38 d2 18 6e c6 33 95 f0 6d f5 72 41

0150 8c 2b 03 e3 98 0b 87 4d a0 25 f6 5f

0160 29 b5 75 c5 eb 1e 42 0a b4 eb 8d 55

0170 6d 7a f2 18 ae ab 9e a4 1f 9e ca 5e

0180 04 94 47 96 8f b9 66 46 09 6b a3 0f

0190 da 1d 5a 36 3f 72 00 21 95 00 b3 37

01a0 bf 64 d7 f7 f2 43 37 a0 ae 09 9d eb

01b0 a4 39 b1 3d 41 f4 5b 9f bf 88 9b 98

01c0 e5 a2 25 a2 5c e7 c2 1e 2a 73 d7 30

01d0 9c 57 9b 03 05 62 21 0a 09 1a d2 6b

01e0 6d 75 de 5d a8 9a 2b 70 1f bb 98 f1

01f0 12 d9 ce a4 06 ec 82 ba b2 69 88 8f

0200 6b b7 57 8b fb 3f de f5 17 43 de b9

0210 ef ba 68 73 01 c7 13 6c ea 05 d9 b9

0220 58 14 9d ad 54 2d 0d b9 fd 8d 12 e7

0230 06 a7 2f 15 5c 5a f1 97 c6 73 f1 a1

Transport Layer Security: Protocol

Packets: 3295 - Displayed: 638 (19.4%)

Profile: Default

Analysis

The Client Hello is the only part of the handshake sent in the clear. The browser lists what it supports — TLS versions, cipher suites, key exchange methods — and the SNI extension reveals the target hostname. After that, the Server Hello picks TLS_AES_256_GCM_SHA384: 256-bit AES encryption with GCM mode and SHA-384 for integrity checking.

From that point on, every Application Data packet in the capture is opaque. You can see the packets exist and their sizes, but the content is unreadable without the session keys. Compare this to the HTTP section — same network, completely different security posture.

The SNI field is an interesting detail. It's sent unencrypted so routers and proxies can route the connection correctly, but it also means anyone watching the network knows which site you're connecting to even over HTTPS. Encrypted Client Hello (ECH) is the newer extension addressing this, though it's not yet universal.

4. Summary

Three protocols, three different relationships with privacy:

- HTTP sends everything in plain text. Headers, payloads, cookies — all visible to anyone on the path.
- DNS resolves domain names before connections open, and by default is also unencrypted — leaking which sites you visit even when the traffic itself is protected.
- TLS negotiates encryption before any application data moves. The handshake is visible; everything after it isn't.

Wireshark makes this concrete in a way that documentation alone can't. Seeing plaintext HTTP headers next to opaque TLS application data in the same capture file explains the difference better than any diagram.

5. Anomaly Comparison — Repeated DNS Queries

The DNS capture shows something beyond a normal single query-response pair. The same domain — `freshshinyshiningmile.neverssl.com` — was queried multiple times in quick succession (packets 259, 260, 261, 273, 274, 275, 276). In normal traffic, a domain is looked up once and the result is cached by the OS or browser. Repeated queries for the same name within seconds suggest the cache either was not used, was cleared, or the TTL was extremely short.

In a real-world security context, repeated DNS queries for the same host can indicate a few things: a misconfigured application retrying failed lookups, DNS-based load balancing with intentionally low TTLs, or in more suspicious scenarios, malware performing repeated beaconing to a command-and-control domain. Recognising this pattern — normal traffic vs. repeated lookups — is one of the practical skills packet analysis develops. The `neverssl.com` capture here is benign, but the pattern is worth knowing.

6. Consolidated Findings

All key values documented across the three captures, as required by Step 6 of the project brief:

Field	HTTP GET	DNS	TLS Handshake
Source IP	10.33.28.192	10.33.28.192	10.33.28.192
Destination IP	34.223.124.45	10.33.28.34 (DNS server)	172.217.22.106 (Google)
Port	80 (HTTP)	53 (UDP)	443 (HTTPS)
Request Method	GET (HTTP/1.1)	N/A	N/A (handshake only)
DNS Resolution	Preceded HTTP request	Resolved in 5.61ms (TX: 0x8bbd)	Preceded TLS connection
TLS Negotiation	None (unencrypted)	None	TLS_AES_256_GCM_SHA384, version TLS 1.2/1.3

Encrypted?	No — plain text	No — plain text	Yes — after handshake
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Capture File

The full packet capture has been saved as a .pcap file (File → Save As in Wireshark) and is available as a supporting reference alongside this report. The .pcap contains all traffic captured during this exercise, including the HTTP, DNS, and TLS sessions documented above.